ECI v7.4 sharp prediction $m_1 = 0$: falsifiability via neutrino mass ordering and future $0\nu\beta\beta$ searches

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The ECI v7.4 framework (NPP20 S_4' modular symmetry with the CSD$(1+\sqrt{6})$ Littlest Modular Seesaw at the fixed point $\tau_S = i$) predicts a rank-2 Majorana mass matrix from a two-right-handed-neutrino seesaw, forcing $m_1 = 0$ exactly in normal ordering. Using NuFIT 6.0 oscillation parameters (arXiv:2410.05380), we compute the effective Majorana mass $m_{\beta\beta} \in [1.50, 3.72]$ meV over all Majorana phases. This window lies *below the sensitivity of all planned neutrinoless double-beta-decay experiments* (KamLAND-Zen, nEXO, LEGEND-1000), but is falsifiable via the neutrino mass ordering: an inverted-ordering determination at $\geq 3\sigma$ by JUNO or DUNE-prec would refute the prediction. We compare with the Tavartkiladze [11] S_3 texture (inverted ordering, $m_{\beta\beta} \gtrsim 34$ meV) and identify mass-ordering determination as the decisive discriminant by 2030–2032.

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INTRODUCTION

The Extended Cosmological Index (ECI) research programme [1] anchors the lepton sector to the LMFDB CM newform of label 4.5.b.a and selects the modular fixed point $\tau_S = i$ via the CM-by-$\mathbb{Q}(i)$ structure. At this fixed point the S_4' weight-2 triplet acquires the alignment [2]

$$Y_3^{(2)}(i) \;\propto\; (1,\; 1+\sqrt{6},\; 1-\sqrt{6}), \tag{1}$$

which defines the CSD$(1+\sqrt{6})$ ansatz in the Littlest Modular Seesaw (LMS) [3]. The two-right-handed-neutrino (2-RHN) structure yields a rank-2 Dirac mass matrix, forcing

$$m_1 = 0 \quad \text{(exact, Normal Ordering)}, \tag{2}$$

with $m_2 = \sqrt{\Delta m^2_{21}}$, $m_3 = \sqrt{\Delta m^2_{31}}$.

The prediction (2) is sharp and structural: it follows from the rank of the seesaw formula, not from a fine-tuning. The question then is whether $m_1 = 0$ is testable. The most direct observable is the effective Majorana mass in neutrinoless double-beta decay ($0\nu\beta\beta$), $m_{\beta\beta} = |m_1|U_{e1}|^2 + m_2|U_{e2}|^2 e^{i\alpha} + m_3|U_{e3}|^2 e^{i\beta}|$. With $m_1 = 0$ this reduces to

$$m_{\beta\beta} = \left| m_2\,|U_{e2}|^2\, e^{i\alpha_{21}} + m_3\,|U_{e3}|^2\, e^{i\alpha_{31}} \right|, \tag{3}$$

where α_{21}, α_{31} are the two Majorana phases, free within ECI v7.4 (the high-energy modular phase does not uniquely pin the low-energy Majorana phases).

NUFIT 6.0 INPUT AND $m_{\beta\beta}$ WINDOW

We use the NuFIT 6.0 global fit [5] (normal-ordering best-fit values, without Super-Kamiokande atmospheric data):

From Table I one derives: $m_2 = 8.654$ meV, $m_3 = 50.130$ meV, $\Sigma m_\nu = 58.78$ meV.

TABLE I. NuFIT 6.0 Normal Ordering best-fit parameters used.

Parameter	Value
$\sin^2\theta_{12}$	0.308
$\sin^2\theta_{13}$	0.02215
Δm^2_{21}	7.49×10^{-5} eV^2
Δm^2_{31}	$+2.513 \times 10^{-3}$ eV^2

The two amplitudes entering Eq. (3) are:

$$A_2 \equiv m_2 \sin^2\theta_{12} \cos^2\theta_{13} = 2.607 \text{ meV}, \tag{4}$$

$$A_3 \equiv m_3 \sin^2\theta_{13} = 1.110 \text{ meV}. \tag{5}$$

The Majorana phase difference $\Delta\phi \equiv \alpha_{31} - \alpha_{21}$ scans $[0, 2\pi)$, giving:

$$\boxed{m_{\beta\beta} \in [1.50,\ 3.72] \text{ meV} \quad \text{(NuFIT 6.0, all Majorana phases)}} \tag{6}$$

with maximum at $\Delta\phi = 0$ ($A_2 + A_3 = 3.72$ meV) and minimum at $\Delta\phi = \pi$ ($|A_2 - A_3| = 1.50$ meV).

EXPERIMENTAL LANDSCAPE

Table II summarises the relevant $0\nu\beta\beta$ experimental program and its sensitivity relative to the ECI prediction (6).

The ECI window [1.50, 3.72] meV is a factor 3–13 below the lower edge of nEXO optimistic sensitivity. To reach $m_{\beta\beta} \sim 3.72$ meV requires $T_{1/2} \gtrsim 2 \times 10^{28}$ yr; to reach $m_{\beta\beta} \sim 1.50$ meV requires $T_{1/2} \gtrsim 10^{29}$ yr (NME-dependent).

A positive signal at $m_{\beta\beta} > 5$ meV in any current-generation experiment would imply $m_1 \neq 0$, *falsifying* the rank-2 structure. A signal at $m_{\beta\beta} > 15$ meV would further require inverted ordering, doubly falsifying ECI v7.4.

TABLE II. $0\nu\beta\beta$ experimental landscape vs. ECI v7.4 prediction $m_{\beta\beta} \in [1.50, 3.72]$ meV. "NO" = not observable; "M" = marginal (optimistic NME).

Experiment	Isotope	$m_{\beta\beta}$ sensitivity	ECI
KamLAND-Zen 2025 [6]	^{136}Xe	28–122 meV	NO
KamLAND-Zen 2027 proj.	^{136}Xe	$\sim$ 12–20 meV	NO
nEXO 2030+ [7]	^{136}Xe	5–20 meV	M
LEGEND-1000 2030+ [8]	^{76}Ge	9–21 meV	NO
Post-2035 ($<$ 2 meV)	TBD	$<$ 2 meV	decisive

INDIRECT FALSIFIABILITY: MASS ORDERING

Since direct $0\nu\beta\beta$ detection is out of reach for near-term experiments, the primary falsifier is the *neutrino mass ordering*:

F8d Inverted ordering at $\geq 3\sigma$ (JUNO 2030, DUNE-prec 2032+). ECI v7.4 demands normal ordering from the 2-RHN seesaw ($m_1 = 0$ is inconsistent with inverted ordering, which requires $m_3 = 0$ or quasi-degenerate spectrum). An IO determination at $\geq 3\sigma$ by JUNO [9] or DUNE [10] *falsifies* the lepton sector of ECI v7.4.

F8a $\delta_{CP} \neq -87^\circ$ at $\geq 3\sigma$ (DUNE 2032+). The LMS at $\tau_S = i$ predicts $\delta_{CP} \approx -87^\circ$. DUNE precision ($\pm 15^\circ$) provides an independent lepton-sector falsifier.

F8b/c $m_{\beta\beta} > 5$ meV detected. Falsifies rank-2 ($m_1 \neq 0$).

Null results at KamLAND-Zen / nEXO / LEGEND-1000 are *consistent* with, but not confirmatory of, the ECI prediction. Confirmation requires either (i) a future experiment reaching $m_{\beta\beta} \sim$ 1–3 meV, or (ii) a joint global fit confirming normal ordering with $m_1 \lesssim 1$ meV at high significance.

COMPARISON WITH TAVARTKILADZE (2025)

The Tavartkiladze 2025 model [11] employs a $\Gamma_2 \simeq S_3$ modular symmetry (level-2, non-holomorphic) with a distinct texture that favours inverted ordering and quasi-degenerate masses, predicting $m_{\beta\beta} \gtrsim 34$ meV.

Table III summarises the key differences.

The two predictions differ by more than an order of magnitude. If JUNO/DUNE establish normal ordering at $\geq 3\sigma$: ECI v7.4 is corroborated, Tavartkiladze constrained. If inverted ordering is established: ECI v7.4 is falsified, Tavartkiladze corroborated. The discriminant is therefore the mass-ordering determination, well within reach by 2030–2032.

TABLE III. ECI v7.4 vs. Tavartkiladze 2025 [11].

Property	ECI v7.4	Tavartkiladze
Modular group	S_4' (level-4)	S_3 (level-2)
Fixed point	$\tau_S = i$	distinct
Ordering	Normal	Inverted
m_1	0 (exact)	non-zero
$m_{\beta\beta}$	$[1.50, 3.72]$ meV	$\gtrsim 34$ meV

DISCUSSION AND CONCLUSION

The ECI v7.4 combination (NPP20 S_4', LMS at $\tau_S = i$, CSD$(1+\sqrt{6})$) yields the sharp prediction $m_1 = 0$ from rank-2 seesaw structure, giving $m_{\beta\beta} \in [1.50, 3.72]$ meV with NuFIT 6.0 parameters. This window is:

- below all near-term $0\nu\beta\beta$ experimental sensitivity;
- consistent with current null results;
- falsifiable by mass-ordering experiments (JUNO/DUNE, 2030–2032);
- qualitatively distinct from the Tavartkiladze IO prediction.

We emphasise that the title "falsifiability via neutrino mass ordering" is precise: the ordering determination is the experimentally accessible falsifier, not direct $0\nu\beta\beta$ detection. A post-2035 experiment reaching $T_{1/2} \sim 10^{29}$ yr would provide the definitive $m_{\beta\beta}$ test.

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